

**KEMPSEY, Australia**WMO: **947850**Lat: **31.0711S**Lon: **152.7717E**Elev: **14**StdP: **101.16**Time Zone: **10.00 (AUS)**Period: **03-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

| Coldest Month | Heating DB |     | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-----|-------------------------------|-----|------|------|-----|------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |     | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99% | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c) | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 7             | 1.6        | 3.0 | -2.4                          | 3.1 | 13.4 | -0.9 | 3.5 | 11.5 | 9.4                   | 16.8 | 8.4 | 16.9 | 1.8                   | 250  | 0.381 |

**Annual Cooling, Dehumidification, and Enthalpy Design Conditions**

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 1             | 10.1                   | 32.6            | 22.4 | 30.8 | 22.3 | 29.4 | 21.8 | 24.2                | 30.2 | 23.4 | 28.8 | 22.7 | 27.6 | 5.1                  | 90   |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 22.4                            | 17.2 | 27.4 | 21.7 | 16.4 | 26.1 | 21.1 | 15.8 | 25.3 | 73.3          | 30.3 | 70.0 | 28.9 | 67.3 | 27.7 | 27.5              |

**Extreme Annual Design Conditions**

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|
| 1%                | 2.5% | 5%  | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |
|                   |      |     | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |
| 8.4               | 7.4  | 6.6 | -1.4                       | 38.2 | 1.3                | 2.9 | -2.4                                               | 40.3 | -3.1       | 42.0 | -3.8       | 43.7 | -4.8       | 45.8 |
|                   |      |     | -1.8                       | 25.9 | 1.3                | 1.0 | -2.7                                               | 26.6 | -3.5       | 27.1 | -4.2       | 27.7 | -5.2       | 28.3 |
|                   |      |     | DB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |
|                   |      |     | WB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |

**Monthly Climatic Design Conditions**

|      |                                                                                  | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |      |
|------|----------------------------------------------------------------------------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|------|
|      |                                                                                  | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |      |
| (6)  | Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 18.0  | 23.3  | 23.0  | 21.7  | 18.8  | 15.3  | 13.4  | 12.3  | 13.1  | 15.8  | 18.0  | 20.2  | 21.8 | (6)  |
| (7)  |                                                                                  | DBStd       | 4.39  | 2.05  | 2.06  | 1.81  | 1.94  | 2.11  | 2.16  | 2.08  | 2.23  | 2.35  | 2.48  | 2.45  | 2.26 | (7)  |
| (8)  |                                                                                  | HDD10.0     | 7     | 0     | 0     | 0     | 0     | 0     | 2     | 4     | 2     | 0     | 0     | 0     | 0    | (8)  |
| (9)  |                                                                                  | HDD18.3     | 739   | 0     | 0     | 1     | 17    | 95    | 147   | 187   | 163   | 83    | 37    | 8     | 2    | (9)  |
| (10) |                                                                                  | CDD10.0     | 2934  | 411   | 363   | 362   | 264   | 165   | 105   | 76    | 98    | 173   | 247   | 306   | 365  | (10) |
| (11) |                                                                                  | CDD18.3     | 624   | 153   | 129   | 104   | 31    | 2     | 0     | 0     | 1     | 6     | 26    | 64    | 108  | (11) |
| (12) |                                                                                  | CDH23.3     | 4547  | 1147  | 880   | 571   | 182   | 35    | 3     | 8     | 27    | 136   | 274   | 503   | 782  | (12) |
| (13) |                                                                                  | CDH26.7     | 1249  | 373   | 270   | 122   | 22    | 1     | 0     | 0     | 5     | 36    | 64    | 137   | 219  | (13) |
| (14) | Wind                                                                             | WSAvg       | 2.7   | 2.9   | 2.8   | 2.6   | 2.5   | 2.3   | 2.6   | 2.6   | 2.8   | 2.9   | 2.9   | 3.0   | 3.0  | (14) |
| (15) | Precipitation                                                                    | PrecAvg     | 1268  | 149   | 148   | 179   | 129   | 112   | 97    | 49    | 48    | 42    | 85    | 102   | 120  | (15) |
| (16) |                                                                                  | PrecMax     | 2264  | 571   | 420   | 594   | 577   | 480   | 558   | 257   | 275   | 143   | 587   | 276   | 354  | (16) |
| (17) |                                                                                  | PrecMin     | 601   | 18    | 32    | 16    | 8     | 3     | 1     | 0     | 0     | 0     | 7     | 24    | 17   | (17) |
| (18) |                                                                                  | PrecStd     | 389   | 115   | 100   | 120   | 136   | 124   | 110   | 65    | 66    | 30    | 102   | 65    | 73   | (18) |
| (19) | Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB    | 34.6  | 34.9  | 31.7  | 29.0  | 25.9  | 23.4  | 24.3  | 26.2  | 31.0  | 31.9  | 33.3  | 33.6 | (19) |
| (20) |                                                                                  |             | MCWB  | 23.2  | 24.6  | 22.9  | 19.9  | 16.8  | 15.1  | 14.3  | 14.2  | 16.8  | 19.1  | 20.4  | 21.8 | (20) |
| (21) |                                                                                  | 2%          | DB    | 32.0  | 31.5  | 29.5  | 26.9  | 24.3  | 21.2  | 21.7  | 23.3  | 26.9  | 28.2  | 30.0  | 30.9 | (21) |
| (22) |                                                                                  |             | MCWB  | 22.9  | 23.5  | 22.5  | 19.4  | 16.6  | 14.6  | 13.6  | 13.7  | 16.5  | 19.2  | 21.2  | 22.3 | (22) |
| (23) |                                                                                  | 5%          | DB    | 30.3  | 29.6  | 27.9  | 25.3  | 22.6  | 20.0  | 20.0  | 21.6  | 24.5  | 26.2  | 28.0  | 29.1 | (23) |
| (24) |                                                                                  |             | MCWB  | 22.7  | 22.6  | 21.6  | 18.7  | 15.9  | 14.0  | 13.0  | 13.5  | 15.8  | 18.6  | 20.6  | 21.5 | (24) |
| (25) |                                                                                  | 10%         | DB    | 28.7  | 28.1  | 26.5  | 24.1  | 21.2  | 18.7  | 18.4  | 20.0  | 22.6  | 24.4  | 26.2  | 27.5 | (25) |
| (26) |                                                                                  |             | MCWB  | 22.1  | 21.6  | 20.7  | 18.3  | 15.0  | 13.6  | 12.3  | 12.8  | 15.2  | 17.7  | 19.7  | 20.7 | (26) |
| (27) | Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB    | 25.3  | 25.6  | 24.1  | 21.5  | 19.2  | 17.2  | 16.8  | 17.2  | 19.4  | 21.8  | 23.7  | 24.5 | (27) |
| (28) |                                                                                  |             | MADB  | 31.7  | 32.3  | 29.7  | 26.3  | 23.6  | 20.2  | 20.2  | 22.8  | 25.5  | 27.9  | 30.0  | 30.8 | (28) |
| (29) |                                                                                  | 2%          | WB    | 24.1  | 24.3  | 23.1  | 20.5  | 17.8  | 16.3  | 15.3  | 15.6  | 18.2  | 20.4  | 22.3  | 23.2 | (29) |
| (30) |                                                                                  |             | MADB  | 30.2  | 30.0  | 28.0  | 24.9  | 21.9  | 19.2  | 18.8  | 20.0  | 23.8  | 25.8  | 28.2  | 28.8 | (30) |
| (31) |                                                                                  | 5%          | WB    | 23.3  | 23.3  | 22.2  | 19.7  | 16.8  | 15.5  | 14.2  | 14.7  | 17.1  | 19.3  | 21.2  | 22.3 | (31) |
| (32) |                                                                                  |             | MADB  | 28.5  | 28.3  | 26.5  | 23.8  | 20.9  | 18.3  | 18.0  | 19.4  | 22.4  | 24.5  | 26.2  | 27.5 | (32) |
| (33) |                                                                                  | 10%         | WB    | 22.6  | 22.4  | 21.5  | 18.9  | 16.0  | 14.6  | 13.3  | 13.7  | 16.1  | 18.4  | 20.3  | 21.6 | (33) |
| (34) |                                                                                  |             | MADB  | 27.2  | 26.6  | 25.4  | 22.8  | 19.7  | 17.4  | 17.0  | 18.2  | 21.0  | 23.0  | 24.7  | 26.1 | (34) |
| (35) | Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR  | 10.1  | 9.8   | 9.7   | 10.9  | 12.9  | 11.6  | 13.7  | 14.9  | 14.2  | 12.5  | 11.2  | 10.7 | (35) |
| (36) |                                                                                  |             | MCDBR | 13.6  | 13.0  | 11.9  | 12.9  | 14.5  | 13.1  | 16.3  | 17.5  | 17.9  | 16.0  | 14.7  | 13.9 | (36) |
| (37) |                                                                                  |             | MCWBR | 6.2   | 5.8   | 5.8   | 6.3   | 7.5   | 7.4   | 8.8   | 9.1   | 8.8   | 8.2   | 7.3   | 6.4  | (37) |
| (38) |                                                                                  | 5% WB       | MCDBR | 11.2  | 11.6  | 11.0  | 11.2  | 12.0  | 9.6   | 12.1  | 14.3  | 15.0  | 13.6  | 12.3  | 11.8 | (38) |
| (39) |                                                                                  |             | MCWBR | 5.4   | 5.5   | 5.6   | 6.0   | 6.8   | 6.1   | 7.3   | 8.3   | 8.2   | 7.5   | 6.5   | 5.9  | (39) |
| (40) | Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.396 | 0.377 | 0.358 | 0.341 | 0.315 | 0.306 | 0.298 | 0.309 | 0.352 | 0.372 | 0.386 | 0.394 | (40) |      |
| (41) |                                                                                  | taud        | 2.470 | 2.547 | 2.590 | 2.604 | 2.629 | 2.647 | 2.654 | 2.608 | 2.476 | 2.457 | 2.470 | 2.464 | (41) |      |
| (42) |                                                                                  | Ebn at Noon | 945   | 944   | 929   | 895   | 873   | 859   | 885   | 917   | 919   | 938   | 948   | 949   | (42) |      |
| (43) |                                                                                  | Edh at Noon | 118   | 106   | 96    | 85    | 74    | 69    | 71    | 83    | 105   | 115   | 117   | 120   | (43) |      |
| (44) | All-Sky Solar<br>Radiation                                                       | RadAvg      | 6.49  | 5.77  | 4.89  | 4.09  | 3.31  | 2.73  | 3.12  | 4.05  | 5.20  | 5.80  | 6.15  | 6.47  | (44) |      |
| (45) |                                                                                  | RadStd      | 0.48  | 0.47  | 0.40  | 0.27  | 0.23  | 0.19  | 0.25  | 0.28  | 0.34  | 0.51  | 0.61  | 0.56  | (45) |      |

**Historical Trends**

|                             | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|-----------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                             |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
| (46) Station Only           | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) Regional (9 neighbors) | +0.45 | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | -55     | +146    | N/A     |

Nomenclature: See separate page